BGP Documentation

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Purpose:

BGP (Border Gateway protocol) is one of the most important routing protocols, as it is used for inter-domain routing across the entire global internet. This lab demonstrates how to configure a network that contains 3 different autonomous systems that are running interior gateway protocols OSPF, EIGRP and IS-IS in a single network, these different interior gateway protocols cannot directly exchange information with each other, since they are designed to only operate on a single autonomous system (AS), this is why BGP is required to act as the inter-AS exchange mechanism by redistributing routes between ASes.

Background:

OSPF (Open Shortest Path First), developed in the 1980s, is an open standard protocol. It is one of the three interior gateway protocols used in this lab. Each router using OSPF shares its own link status using something called Link-State Advertisements (LSA), and stores the info in a shared database, building a complete map of the network by connecting interfaces to a central backbone (Area 0), with each area having its own Link-state database (LSDB) storing information from LSAs, thus enabling scaling.

IS-IS (Intermediate System to Intermediate system), also developed in the late 1980s by Digital Equipment Corporation (DEC) is another link-state interior gateway protocol used in this lab, which means it also builds a map of the network. Unlike OSPF IS-IS runs directly on the Data Link Layer, building a hierarchical chain, with routers being in different levels instead of areas. Routers get categorized into Level 1: Routing within a local area, and Level 2: Routing between areas (the backbone), this leveling system makes IS-IS highly scalable, making it commonly used by Service Providers.

EIGRP (Enhanced Interior Gateway Protocol), is the third IGP used in this lab, it is an advanced distance routing protocol that replaced Interior Gateway Protocol (IGRP). It's a cisco proprietary protocol, meaning it is commonly only available on cisco routers. EIGRP shares routes with other routers in the same AS, EIGRP only sends incremental updates, reducing workload on the router. EIGRP also utilizes DUAL (Diffusing Update Algorithm) for loop free paths.

BGP (Border Gateway Protocol): January 1989, at the 12th IETF meeting in Austin, Texas, Yakov Rekhter, Len Bosack(One of the founders of cisco) and Kirk Lougheed recorded a design on 2 napkins for BGP. Over the next years BGP went through 4 versions, with the most recent version being BGP4, developed in 1994 with it being used on the internet since. The internet is made of over 70,000 autonomous systems being made of networks run by ISPs, tech companies, and universities. BGP acts like the glue of the internet, being the only protocol that allows all these ASes to be able to talk to each other.

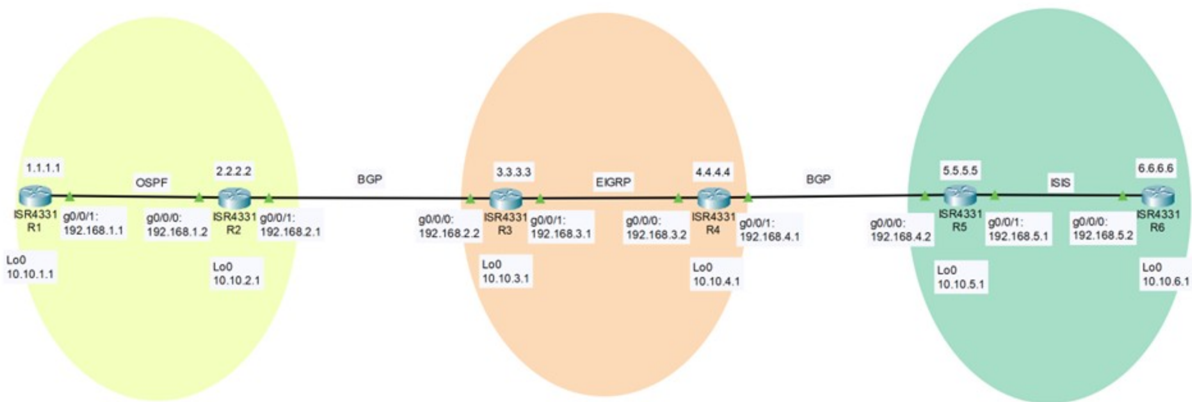
BGP is an exterior gateway protocol made to exchange routing information between ASes. BGP for within a single AS is called iBGP (Interior border gateway protocol), whereas in this lab, EBGP (Exterior Border Gateway Protocol) is used to connect the OSPF, IS-IS and EIGRP ASes. Each IGP runs independently, routers running OSPF have their own database, EIGRP router have theirs and IS-IS also has a different one. Without BGP, each AS within the network using a different IGP wouldn't know each other's routes. On the router that borders each

AS, mutual redistribution needs to be configured, border routers take specific internal routes and advertises them to the entire network, outside its AS, the border router running BGP also takes global routes from other ASes and puts them in their own internal database, letting OSPF, EIGRP and IS-IS communicate with each other and obtain each other's routes.

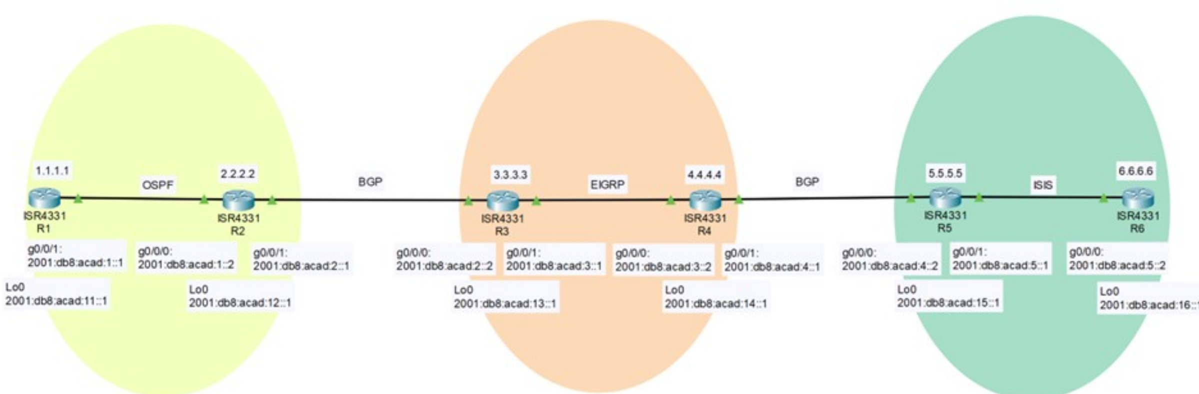
When routes go from protocol to protocol, normally it wouldn't work since for example, a router using EIGRP would have no idea what a "Cost" or a OSPF Link-State advertisement is, this is why BGP is needed, when mutual redistribution is configured on a border router, BGP gets rid of the local context like the LSA type of OSPF or K-values of EIGRP leaving just the Prefix. Then the route is repackaged into a BGP Update message, this message is made into a universal set of path attributes, letting any router be able to understand the route's history and priority.

Configurations:

IPv4 Topology:



IPv6 Topology:



Router Configurations:

OSPF configurations:

```
Rx(config)#router ospf (process-ID)
Rx(config-router)#router-id (router-ID)
Rx(config-router)#exit
Rx(config)#int g0/0/x
Rx(config-if)#ip address (IPv4 address) (subnet mask)
Rx(config-if)#ip ospf (process-ID) area (area)
Rx(config-if)#no shutdown
Rx(config-if)#exit
Rx(config)#int lo0
Rx(config-if)#ip address (IPv4 address) (subnet mask)
Rx(config-if)#ip ospf (process-ID) area (area)
Rx(config-if)#no shutdown
Rx(config-if)#end
Rx#clear ip ospf process
Reset selected OPSFv2 processes? [no]:yes
```

OSPFV3 configurations:

```
Rx(config)#ipv6 router ospf (process-ID)
Rx(config-router)#router-id (router-ID)
Rx(config-router)#exit
Rx(config)#int g0/0/x
Rx(config-if)#ipv6 address (IPv6 address)
Rx(config-if)#ipv6 ospf (process-ID) area (area)
Rx(config-if)#no shutdown
Rx(config-if)#exit
Rx(config)#int lo0
Rx(config-if)#ipv6 address (IPv6 address)
Rx(config-if)#ip ospf (process-ID) area (area)
Rx(config-if)#no shutdown
Rx(config-if)#end
Rx#clear ipv6 ospf process
Reset selected OPSFv3 processes? [no]:yes

Rx(config)#ip router eigrp (process-ID)
Rx(config-router)#eigrp router-id (router-ID)
Rx(config-router)#default-metric (metrics)
Rx(config-router)#network (IPv4 address) (Wildcard mask)
Rx(config-router)#network (IPv4 address) (Wildcard mask)
Rx(config-router)#exit
Rx(config-if)#end
```

EIGRPv4 configurations:

```
Rx(config)#ip router eigrp (process-ID)
Rx(config-router)#eigrp router-id (router-ID)
Rx(config-router)#default-metric (metrics)
Rx(config-router)#network (IPv4 address) (Wildcard mask)
Rx(config-router)#network (IPv4 address) (Wildcard mask)
Rx(config-router)#exit
Rx(config-if)#end
```

EIGRPv6 Configuration:

```
Rx(config)#ipv6 router eigrp (process-ID)
Rx(config-router)#eigrp router-id (router-ID)
Rx(config-router)#default-metric (metrics)
Rx(config-router)#exit
Rx(config)#int g0/0/x
Rx(config-if)#ipv6 address (IPv6 address)
Rx(config-if)#ipv6 enable
Rx(config-if)#ipv6 eigrp (process-ID)
Rx(config-if)#no shutdown
Rx(config-if)#exit
Rx(config)#int lo0
Rx(config-if)#ipv6 address (IPv6 address)
Rx(config-if)#ipv6 enable
Rx(config-if)#ipv6 eigrp (process-ID)
Rx(config-if)#no shutdown
Rx(config-if)#end
```

IS-IS IPv4 Configuration:

```
Rx(config)#router isis (process-ID)
Rx(config-router)#net (NET address)
Rx(config-router)#exit
Rx(config)#int g0/0/x
Rx(config-if)#ip address (IPv4 address) (subnet mask)
Rx(config-if)#ip router isis (process-ID)
Rx(config-if)#no shutdown
Rx(config-if)#exit
Rx(config)#int lo0
Rx(config-if)#ip address (IPv4 address) (subnet mask)
Rx(config-if)#ip router isis (process-ID)
Rx(config-if)#no shutdown
Rx(config-if)#end
```

```
IS-IS IPv6 Configuration:
Rx(config)#router isis (process-ID)
```

```
Rx(config-router)#net (NET address)
Rx(config-router)#exit
Rx(config)#int g0/0/x
Rx(config-if)#ipv6 address (IPv6 address)
Rx(config-if)#ipv6 enable
Rx(config-if)#ipv6 router isis (process-ID)
Rx(config-if)#no shutdown
Rx(config-if)#exit
Rx(config)#int lo0
Rx(config-if)#ipv6 address (IPv6 address)
Rx(config-if)#ipv6 enable
Rx(config-if)#ipv6 router isis (process-ID)
Rx(config-if)#no shutdown
Rx(config-if)#end
```

BGP (IPv4: OSPF to BGP):

```
Example Router Configuration:
Rx(config)#router bgp (ASN)
Rx(config-router)#address-family ipv4
Rx(config-router-ad)#neighbor (IPv4 address) remote-as (ASN)
Rx(config-router-ad)#neighbor (IPv4 address) activate
Rx(config-router-ad)#redistribute (IGP Protocol) (process-ID)
include-connected
Rx(config-router-ad)#end
OSPF
Rx(config)#router ospf (Process-ID)
Rx(config-router)#redistribute bgp (ASN) include-connected
Rx(config-router)#end
IS-IS
Rx(config)#router isis (Process-ID)
Rx(config-router)#address-family ipv4
Rx(config-router-ad)#redistribute bgp (ASN) include-connected
Rx(config-router-ad)#end
EIGRP
Rx(config)#router eigrp (Process-ID)
Rx(config-router)#redistribute bgp (ASN) include-connected
metric (bandwidth, delay, reliability, load, MTU)
Rx(config-router)#end
```

Network Configurations:

R1

```
hostname R1

no ip domain lookup
```

```
interface Loopback0
 ip address 10.10.1.1 255.255.255.0
 ip ospf 1 area 2
 ipv6 address 2001:DB8:ACAD:11::1/64
 ipv6 ospf 1 area 2

interface GigabitEthernet0/0/1
 ip address 192.168.1.1 255.255.255.0
 ip ospf 1 area 0
 negotiation auto
 ipv6 address FE80::1 link-local
 ipv6 address 2001:DB8:ACAD:1::1/64
 ipv6 ospf 1 area 0

router ospfv3 1
 router-id 1.1.1.1

router ospf 1
 router-id 1.1.1.1
 network 192.168.1.0 0.0.0.255 area 0

end
```

R2

```
hostname R2

no ip domain lookup

ipv6 unicast-routing

interface Loopback0
 ip address 10.10.2.1 255.255.255.0
 ip ospf 1 area 1
 ipv6 address 2001:DB8:ACAD:12::1/64
 ipv6 ospf 1 area 1

interface GigabitEthernet0/0/0
 ip address 192.168.1.2 255.255.255.0
 ip ospf 1 area 0
 ipv6 address FE80::2 link-local
 ipv6 address 2001:DB8:ACAD:1::2/64
 ipv6 ospf 1 area 0

interface GigabitEthernet0/0/1
 ip address 192.168.2.1 255.255.255.0
```

```
 ipv6 address FE80::1 link-local
 ipv6 address 2001:DB8:ACAD:2::1/64

router ospfv3 1
 router-id 2.2.2.2


router ospf 1
 router-id 2.2.2.2
 redistribute connected subnets
 redistribute bgp 65001 subnets
 network 192.168.1.0 0.0.0.255 area 0

router bgp 65001
 bgp log-neighbor-changes
 no bgp default ipv4-unicast
 neighbor 2001:DB8:ACAD:2::2 remote-as 65002
 neighbor 192.168.2.2 remote-as 65002

 address-family ipv4
  redistribute connected
  redistribute ospf 1 include-connected
  neighbor 192.168.2.2 activate
 exit-address-family

 address-family ipv6
  redistribute connected
  redistribute ospfv3 1 include-connected
  neighbor 2001:DB8:ACAD:2::2 activate
 exit-address-family

end
```

R3

```
hostname R3

no ip domain lookup

ipv6 unicast-routing

interface Loopback0
 ip address 10.10.3.1 255.255.255.0
 ipv6 address 2001:DB8:ACAD:13::1/64
 ipv6 enable
 ipv6 eigrp 100
```

```
interface GigabitEthernet0/0/0
 ip address 192.168.2.2 255.255.255.0
 ipv6 address FE80::2 link-local
 ipv6 address 2001:DB8:ACAD:2::2/64

interface GigabitEthernet0/0/1
 ip address 192.168.3.1 255.255.255.0
 ipv6 address FE80::1 link-local
 ipv6 address 2001:DB8:ACAD:3::1/64
 ipv6 enable
 ipv6 eigrp 100

router eigrp 100
 default-metric 10000 100 255 1 1500
 network 10.0.0.0
 network 192.168.3.0
 redistribute bgp 65002 metric 10000 100 255 1 1500
 redistribute ospf 1
 redistribute connected metric 10000 100 255 1 1500
 eigrp router-id 3.3.3.3

router bgp 65002
 bgp log-neighbor-changes
 no bgp default ipv4-unicast
 neighbor 2001:DB8:ACAD:2::1 remote-as 65001
 neighbor 192.168.2.1 remote-as 65001
 neighbor 192.168.4.2 remote-as 65003

 address-family ipv4
  redistribute eigrp 100
  neighbor 192.168.2.1 activate
  neighbor 192.168.4.2 activate
 exit-address-family

 address-family ipv6
  redistribute eigrp 100 include-connected
  neighbor 2001:DB8:ACAD:2::1 activate
 exit-address-family

ipv6 router eigrp 100
 eigrp router-id 3.3.3.3
 redistribute bgp 65002 metric 10000 100 255 1 1500
 redistribute connected

end
```

R4

```
hostname R4

no ip domain lookup

ipv6 unicast-routing

interface Loopback0
 ip address 10.10.4.1 255.255.255.0
 ipv6 address 2001:DB8:ACAD:14::1/64
 ipv6 enable
 ipv6 eigrp 100

interface GigabitEthernet0/0/0
 ip address 192.168.3.2 255.255.255.0
 ipv6 address FE80::2 link-local
 ipv6 address 2001:DB8:ACAD:3::2/64
 ipv6 enable
 ipv6 eigrp 100

interface GigabitEthernet0/0/1
 ip address 192.168.4.1 255.255.255.0
 ipv6 address FE80::1 link-local
 ipv6 address 2001:DB8:ACAD:4::1/64

router eigrp 100
 network 10.0.0.0
 network 192.168.3.0
 network 192.168.4.0
 redistribute bgp 65002 metric 100000 100 255 1 1500
 redistribute connected metric 10000 100 255 1 1500
 eigrp router-id 4.4.4.4

router bgp 65002
 bgp log-neighbor-changes
 no bgp default ipv4-unicast
 neighbor 2001:DB8:ACAD:2::1 remote-as 65001
 neighbor 2001:DB8:ACAD:4::2 remote-as 65003
 neighbor 192.168.2.1 remote-as 65001
 neighbor 192.168.4.2 remote-as 65003

 address-family ipv4
  redistribute connected
  redistribute eigrp 100
  neighbor 192.168.2.1 activate
  neighbor 192.168.4.2 activate
 exit-address-family
```

```
 address-family ipv6
  redistribute connected
  redistribute eigrp 100 include-connected
  neighbor 2001:DB8:ACAD:2::1 activate
  neighbor 2001:DB8:ACAD:4::2 activate
 exit-address-family

ipv6 router eigrp 100
 eigrp router-id 4.4.4.4
 redistribute bgp 65002 metric 10000 100 255 1 1500
 redistribute connected metric 10000 100 255 1 1500

end
```

R5

```
hostname R5

no ip domain lookup

ipv6 unicast-routing

interface Loopback0
 ip address 10.10.5.1 255.255.255.0
 ip router isis 1
 ipv6 address 2001:DB8:ACAD:15::1/64
 ipv6 router isis 1

interface GigabitEthernet0/0/0
 ip address 192.168.4.2 255.255.255.0
 ipv6 address FE80::2 link-local
 ipv6 address 2001:DB8:ACAD:4::2/64

interface GigabitEthernet0/0/1
 ip address 192.168.5.1 255.255.255.0
 ip router isis 1
 ipv6 address FE80::1 link-local
 ipv6 address 2001:DB8:ACAD:5::1/64
 ipv6 router isis 1

router isis 1
 net 49.0001.0000.0000.000a.00

 address-family ipv4
  redistribute connected
  redistribute bgp 65003
```

```
 exit-address-family

 address-family ipv6
  redistribute connected
  redistribute bgp 65003
 exit-address-family

router bgp 65003
 bgp log-neighbor-changes
 neighbor 2001:DB8:ACAD:4::1 remote-as 65002
 neighbor 192.168.4.1 remote-as 65002

 address-family ipv4
  redistribute connected
  redistribute isis 1 level-1
  no neighbor 2001:DB8:ACAD:4::1 activate
  neighbor 192.168.4.1 activate
 exit-address-family

 address-family ipv6
  redistribute isis 1 level-1-2 include-connected
  redistribute isis level-1
  neighbor 2001:DB8:ACAD:4::1 activate
 exit-address-family

end
```

R6

```
hostname R6

no aaa new-model

no ip domain lookup

ipv6 unicast-routing

interface Loopback0
 ip address 10.10.6.1 255.255.255.0
 ip router isis 1
 ipv6 address 2001:DB8:ACAD:16::1/64
 ipv6 router isis 1

interface GigabitEthernet0/0/0
 ip address 192.168.5.2 255.255.255.0
 ip router isis 1
 ipv6 address FE80::2 link-local
```

```
 ipv6 address 2001:DB8:ACAD:5::2/64
 ipv6 router isis 1

router isis 1
 net 49.0001.0000.0000.000b.00

end
```

Proof of Connectivity

Pings across network:

IPv4: R1 to R6 and R6 to R1

```
R1#ping 10.10.6.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.6.1, timeout is 2
seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max =
1/3/13 ms
R1#traceroute 10.10.6.1
Type escape sequence to abort.
Tracing the route to 10.10.6.1
VRF info: (vrf in name/id, vrf out name/id)
  1 192.168.1.2 1 msec 1 msec 1 msec
  2 192.168.2.2 1 msec 0 msec 1 msec
  3 192.168.3.2 1 msec 1 msec 1 msec
  4 192.168.4.2 1 msec 1 msec 1 msec
  5 192.168.5.2 2 msec 2 msec *

R6#ping 10.10.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.1.1, timeout is 2
seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max =
1/1/2 ms
R6#traceroute 10.10.1.1
Type escape sequence to abort.
Tracing the route to 10.10.1.1
VRF info: (vrf in name/id, vrf out name/id)
  1 192.168.5.1 1 msec 0 msec 1 msec
  2 192.168.4.1 1 msec 1 msec 1 msec
  3 192.168.3.1 0 msec 1 msec 1 msec
  4 192.168.2.1 1 msec 2 msec 1 msec
  5 192.168.1.1 1 msec *  1 msec
```

IPv6: R1 to R6 and R6 to R1

```
R1#ping 2001:db8:acad:16::1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:16::1, timeout
is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max =
1/2/6 ms
R1#traceroute 2001:db8:acad:16::1
Type escape sequence to abort.
Tracing the route to 2001:DB8:ACAD:16::1

  1 2001:DB8:ACAD:1::2 2 msec 1 msec 0 msec
  2 2001:DB8:ACAD:2::2 0 msec 1 msec 1 msec
  3 2001:DB8:ACAD:3::2 1 msec 2 msec 1 msec
  4 2001:DB8:ACAD:4::2 1 msec 2 msec 1 msec
  5 2001:DB8:ACAD:5::2 1 msec 2 msec 1 msec
```

```
R6#ping 2001:db8:acad:11::1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:11::1, timeout
is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max =
1/2/6 ms
R6#traceroute 2001:db8:acad:11::1
Type escape sequence to abort.
Tracing the route to 2001:DB8:ACAD:11::1

  1 2001:DB8:ACAD:5::1 1 msec 0 msec 0 msec
  2 2001:DB8:ACAD:4::1 1 msec 1 msec 1 msec
  3 2001:DB8:ACAD:3::1 1 msec 1 msec 1 msec
  4 2001:DB8:ACAD:2::1 1 msec 2 msec 1 msec
  5 2001:DB8:ACAD:1::1 1 msec 1 msec 2 msec
```

Proof of BGP
IPv4: BGP Database

```
R3#show ip bgp
BGP table version is 67, local router ID is 192.168.2.2
Status codes: s suppressed, d damped, h history, * valid, >
best, i - internal,
```

```
              r RIB-failure, S Stale, m multipath, b backup-
path, f RT-Filter,
              x best-external, a additional-path, c RIB-
compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight
Path
 *>  10.10.1.1/32     192.168.2.1              2             0
65001 ?
 *>  10.10.2.0/24     192.168.2.1              0             0
65001 ?
 *>  10.10.3.0/24     0.0.0.0                  0         32768 ?
 *>  10.10.4.0/24     192.168.3.2         130816         32768 ?
 *>  10.10.5.0/24     192.168.3.2          51456         32768 ?
 *>  10.10.6.0/24     192.168.3.2          51456         32768 ?
 *>  192.168.1.0      192.168.2.1              0             0
65001 ?
 r>  192.168.2.0      192.168.2.1              0             0
65001 i
 *>  192.168.3.0      0.0.0.0                  0         32768 ?
 *>  192.168.4.0      192.168.3.2           3072         32768 ?
 *>  192.168.5.0      192.168.3.2          51456         32768 ?
```

IPv4: BGP Routes

```
R4#show ip route bgp
Codes: L - local, C - connected, S - static, R - RIP, M -
mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, m -
OMP
       n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 -
IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-
user static route
       H - NHRP, G - NHRP registered, g - NHRP registration
summary
       o - ODR, P - periodic downloaded static route, l - LISP
       a - application route
       + - replicated route, % - next hop override, p -
overrides from PfR
```

```
       & - replicated local route overrides by connected

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
B        10.10.5.0/24 [20/0] via 192.168.4.2, 1d00h
B        10.10.6.0/24 [20/20] via 192.168.4.2, 1d00h
B     192.168.5.0/24 [20/0] via 192.168.4.2, 1d00h
```

IPv6: BGP Database

```
R3#show bgp ipv6 unicast
BGP table version is 503, local router ID is 192.168.2.2
Status codes: s suppressed, d damped, h history, * valid, >
best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-
path, f RT-Filter,
              x best-external, a additional-path, c RIB-
compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight
Path
 *>  2001:DB8:ACAD:1::/64
                       2001:DB8:ACAD:2::1
                                                 0             0
65001 ?
 r>  2001:DB8:ACAD:2::/64
                       2001:DB8:ACAD:2::1
                                                 0             0
65001 ?
 *>  2001:DB8:ACAD:3::/64
                       ::                        0
32768 ?
 *>  2001:DB8:ACAD:4::/64
                       FE80::2              281856
32768 ?
 *>  2001:DB8:ACAD:5::/64
                       FE80::2              281856
32768 ?
 *>  2001:DB8:ACAD:11::1/128
                       2001:DB8:ACAD:2::1
     Network          Next Hop            Metric LocPrf Weight
Path
                                                 1             0
65001 ?
```

```
 *>  2001:DB8:ACAD:12::/64
                       2001:DB8:ACAD:2::1
                                                        0              0
65001 ?
 *>  2001:DB8:ACAD:13::/64
                       ::                               0
32768 ?
 *>  2001:DB8:ACAD:14::/64
                       FE80::2                     130816
32768 ?
 *>  2001:DB8:ACAD:15::/64
                       FE80::2                     281856
32768 ?
 *>  2001:DB8:ACAD:16::/64
                       FE80::2                     281856
32768 ?
```

IPv6: BGP Routes

```
R4#show ipv6 route bgp
IPv6 Routing Table - default - 15 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static
route
       B - BGP, R - RIP, H - NHRP, I1 - ISIS L1
       I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D -
EIGRP
       EX - EIGRP external, ND - ND Default, NDp - ND Prefix,
DCE - Destination
       NDr - Redirect, O - OSPF Intra, OI - OSPF Inter, OE1 -
OSPF ext 1
       OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA
ext 2
       a - Application, m - OMP
B   2001:DB8:ACAD:5::/64 [20/0], tag 65003
     via FE80::2, GigabitEthernet0/0/1
B   2001:DB8:ACAD:15::/64 [20/0], tag 65003
     via FE80::2, GigabitEthernet0/0/1
B   2001:DB8:ACAD:16::/64 [20/20], tag 65003
     via FE80::2, GigabitEthernet0/0/1
```

Proof of EIGRP

IPv4: EIGRP Neighbors

```
R4#show ip eigrp neighbors
```

```
EIGRP-IPv4 Neighbors for AS(100)
H   Address                 Interface              Hold Uptime
SRTT   RTO  Q  Seq
                                                   (sec)
(ms)       Cnt Num
0   192.168.3.1             Gi0/0/0                  11 1w0d
1   100  0  51
```

IPv4: EIGRP routes

```
R4#show ip route eigrp
Codes: L - local, C - connected, S - static, R - RIP, M -
mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, m -
OMP
       n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 -
IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-
user static route
       H - NHRP, G - NHRP registered, g - NHRP registration
summary
       o - ODR, P - periodic downloaded static route, l - LISP
       a - application route
       + - replicated route, % - next hop override, p -
overrides from PfR
       & - replicated local route overrides by connected

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
D EX     10.10.1.1/32
           [170/281856] via 192.168.3.1, 00:20:36,
GigabitEthernet0/0/0
D EX     10.10.2.0/24
           [170/281856] via 192.168.3.1, 4d01h,
GigabitEthernet0/0/0
D        10.10.3.0/24 [90/130816] via 192.168.3.1, 4d02h,
GigabitEthernet0/0/0
D EX  192.168.1.0/24
           [170/281856] via 192.168.3.1, 00:21:07,
GigabitEthernet0/0/0
```

```
D EX  192.168.2.0/24 [170/281856] via 192.168.3.1, 4d01h,
GigabitEthernet0/0/0
```

IPv6: EIGRP Neighbors

```
R4#show ipv6 eigrp neighbors
EIGRP-IPv6 Neighbors for AS(100)
H   Address                 Interface              Hold Uptime
SRTT   RTO  Q  Seq
                                                   (sec)
(ms)       Cnt Num
0   Link-local address:     Gi0/0/0                  13 1d00h
1   100  0  17
    FE80::1
```

IPv6: EIGRP routes

```
R4#show ipv6 route eigrp
IPv6 Routing Table - default - 15 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static
route
       B - BGP, R - RIP, H - NHRP, I1 - ISIS L1
       I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D -
EIGRP
       EX - EIGRP external, ND - ND Default, NDp - ND Prefix,
DCE - Destination
       NDr - Redirect, O - OSPF Intra, OI - OSPF Inter, OE1 -
OSPF ext 1
       OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA
ext 2
       a - Application, m - OMP
EX  2001:DB8:ACAD:1::/64 [170/281856]
     via FE80::1, GigabitEthernet0/0/0
EX  2001:DB8:ACAD:2::/64 [170/3072]
     via FE80::1, GigabitEthernet0/0/0
EX  2001:DB8:ACAD:11::1/128 [170/281856]
     via FE80::1, GigabitEthernet0/0/0
EX  2001:DB8:ACAD:12::/64 [170/281856]
     via FE80::1, GigabitEthernet0/0/0
D   2001:DB8:ACAD:13::/64 [90/130816]
     via FE80::1, GigabitEthernet0/0/0
```

Proof of IS-IS

IS-IS: CLNS neighbors

```
R5#show clns neighbors

Tag 1:
System Id       Interface     SNPA                State
Holdtime  Type Protocol
R6              Gi0/0/1       0027.90d5.fa40      Up     28
L1L2 IS-IS

R6#show clns neighbors
Tag 1:
System Id       Interface     SNPA                State
Holdtime  Type Protocol
R5              Gi0/0/0       0027.90d5.fad1      Up     9
L1L2 IS-IS
```

IPv4: IS-IS routes

```
R6#show ip route isis
Codes: L - local, C - connected, S - static, R - RIP, M -
mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 -
IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-
user static route
       o - ODR, P - periodic downloaded static route, H - NHRP,
l - LISP
       a - application route
       + - replicated route, % - next hop override, p -
overrides from PfR

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
i L2     10.10.1.1/32 [115/10] via 192.168.5.1, 00:30:12,
GigabitEthernet0/0/0
i L2     10.10.2.0/24 [115/10] via 192.168.5.1, 1d00h,
GigabitEthernet0/0/0
i L2     10.10.3.0/24 [115/10] via 192.168.5.1, 1d00h,
GigabitEthernet0/0/0
i L2     10.10.4.0/24 [115/10] via 192.168.5.1, 1d00h,
GigabitEthernet0/0/0
```

```
i L1     10.10.5.0/24 [115/20] via 192.168.5.1, 1d01h,
GigabitEthernet0/0/0
i L2  192.168.1.0/24 [115/10] via 192.168.5.1, 00:30:43,
GigabitEthernet0/0/0
i L2  192.168.2.0/24 [115/10] via 192.168.5.1, 1d00h,
GigabitEthernet0/0/0
i L2  192.168.3.0/24 [115/10] via 192.168.5.1, 1d00h,
GigabitEthernet0/0/0
i L2  192.168.4.0/24 [115/10] via 192.168.5.1, 1d00h,
GigabitEthernet0/0/0
```

IPv6: IS-IS routes

```
R6#show ipv6 route isis
IPv6 Routing Table - default - 14 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static
route
       B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2
       IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX -
EIGRP external
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr
- Redirect
       O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 -
OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a -
Application
I2  2001:DB8:ACAD:1::/64 [115/10]
     via FE80::1, GigabitEthernet0/0/0
I2  2001:DB8:ACAD:2::/64 [115/10]
     via FE80::1, GigabitEthernet0/0/0
I2  2001:DB8:ACAD:3::/64 [115/10]
     via FE80::1, GigabitEthernet0/0/0
I2  2001:DB8:ACAD:4::/64 [115/10]
     via FE80::1, GigabitEthernet0/0/0
I2  2001:DB8:ACAD:11::1/128 [115/10]
     via FE80::1, GigabitEthernet0/0/0
I2  2001:DB8:ACAD:12::/64 [115/10]
     via FE80::1, GigabitEthernet0/0/0
I2  2001:DB8:ACAD:13::/64 [115/10]
     via FE80::1, GigabitEthernet0/0/0
I2  2001:DB8:ACAD:14::/64 [115/10]
     via FE80::1, GigabitEthernet0/0/0
I1  2001:DB8:ACAD:15::/64 [115/20]
     via FE80::1, GigabitEthernet0/0/0
```

Proof of OSPF

IPv4: OSPF neighbors

```
R1#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address
Interface
2.2.2.2           1   FULL/DR         00:00:31    192.168.1.2
GigabitEthernet0/0/1
```

IPv4: OSPF database

```
R2#show ip ospf database

            OSPF Router with ID (2.2.2.2) (Process ID 1)

                Router Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum
Link count
1.1.1.1         1.1.1.1         983         0x8000014B 0x00FD09
1
2.2.2.2         2.2.2.2         974         0x800000B7 0x00EFA0
1

                Net Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum
192.168.1.2     2.2.2.2         990         0x800000B0 0x00A960

                Summary Net Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum
10.10.1.1       1.1.1.1         1031        0x800000B0 0x00066C
10.10.2.1       2.2.2.2         1030        0x80000132 0x00D614

                Router Link States (Area 1)

Link ID         ADV Router      Age         Seq#       Checksum
Link count
2.2.2.2         2.2.2.2         1030        0x800000B2 0x00BA9B
1

                Summary Net Link States (Area 1)

Link ID         ADV Router      Age         Seq#       Checksum
10.10.1.1       2.2.2.2         990         0x80000001 0x0051CB
```

```
192.168.1.0     2.2.2.2         1030        0x800000B0 0x003BDE

                Type-5 AS External Link States

Link ID         ADV Router      Age         Seq#       Checksum
Tag
10.10.3.0       2.2.2.2         145         0x800000B0 0x00D328
65002
10.10.4.0       2.2.2.2         145         0x80000107 0x00198A
65002
10.10.5.0       2.2.2.2         402         0x8000002C 0x00C6B7
65002
10.10.6.0       2.2.2.2         402         0x8000002C 0x00BBC1
65002
192.168.2.0     2.2.2.2         145         0x80000132 0x007E7C
0
192.168.3.0     2.2.2.2         145         0x80000132 0x00170D
65002
192.168.4.0     2.2.2.2         402         0x8000002C 0x001B0F
65002
192.168.5.0     2.2.2.2         402         0x8000002C 0x001019
65002
```

IPv4: OSPF routes

```
R2#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M -
mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 -
IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-
user static route
       o - ODR, P - periodic downloaded static route, H - NHRP,
l - LISP
       a - application route
       + - replicated route, % - next hop override, p -
overrides from PfR

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
```

```
O IA     10.10.1.1/32 [110/2] via 192.168.1.1, 00:16:59,
GigabitEthernet0/0/0
```

IPv6: OSPF neighbors

```
R1#show ipv6 ospf neighbor

            OSPFv3 Router with ID (1.1.1.1) (Process ID 1)

Neighbor ID     Pri   State           Dead Time   Interface ID
Interface
2.2.2.2           1   FULL/DR         00:00:32    6
GigabitEthernet0/0/1
```

IPv6: OSPF database

```
R2#show ipv6 ospf database

            OSPFv3 Router with ID (2.2.2.2) (Process ID 1)

                Router Link States (Area 0)

ADV Router       Age         Seq#        Fragment ID  Link count
Bits
 1.1.1.1         1267        0x8000014C  0            1
B
 2.2.2.2         1539        0x800000BC  0            1
B E

                Net Link States (Area 0)

ADV Router       Age         Seq#        Link ID    Rtr count
 2.2.2.2         1556        0x80000001  6          2

                Inter Area Prefix Link States (Area 0)

ADV Router       Age         Seq#        Prefix
 1.1.1.1         1267        0x800000B0  2001:DB8:ACAD:11::1/128
 2.2.2.2         1130        0x800000B1  2001:DB8:ACAD:12::1/128

                Link (Type-8) Link States (Area 0)

ADV Router       Age         Seq#        Link ID    Interface
 1.1.1.1         1267        0x800000B2  7          Gi0/0/0
 2.2.2.2         1130        0x800000B3  6          Gi0/0/0

                Intra Area Prefix Link States (Area 0)
```

```
ADV Router        Age         Seq#        Link ID    Ref-lstype
Ref-LSID
 2.2.2.2          1556        0x80000001  6144       0x2002
6

                Router Link States (Area 1)

ADV Router        Age         Seq#        Fragment ID  Link count
Bits
 2.2.2.2          1596        0x800000B3  0             0
B E

                Inter Area Prefix Link States (Area 1)

ADV Router        Age         Seq#        Prefix
 2.2.2.2          1130        0x800000B0  2001:DB8:ACAD:1::/64
 2.2.2.2          1556        0x80000001  2001:DB8:ACAD:11::1/128

                Intra Area Prefix Link States (Area 1)

ADV Router        Age         Seq#        Link ID    Ref-lstype
Ref-LSID
 2.2.2.2          1130        0x800000AF  0          0x2001
0

                Type-5 AS External Link States

ADV Router        Age         Seq#        Prefix
 2.2.2.2          625         0x80000002  2001:DB8:ACAD:3::/64
 2.2.2.2          625         0x80000002  2001:DB8:ACAD:5::/64
 2.2.2.2          625         0x80000002  2001:DB8:ACAD:13::/64
 2.2.2.2          625         0x80000002  2001:DB8:ACAD:14::/64
 2.2.2.2          625         0x80000002  2001:DB8:ACAD:15::/64
 2.2.2.2          625         0x80000002  2001:DB8:ACAD:16::/64
 2.2.2.2          368         0x80000002  2001:DB8:ACAD:4::/64
```

IPv6: OSPF routes

```
R2#show ipv6 route ospf
IPv6 Routing Table - default - 15 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static
route
       B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2
       IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX -
EIGRP external
```

```
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr
- Redirect
       O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 -
OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a -
Application
OI  2001:DB8:ACAD:11::1/128 [110/1]
     via FE80::1, GigabitEthernet0/0/0
```

Problems:

In the lab, redistribution of the EIGRP requires more than redistribution of IS-IS or OSPF, this is because IS-IS and OSPF use metrics as cost, while EIGRP uses a different set of metrics (bandwidth, delay, reliability, load), so using the command `redistribute bgp 65001 include-connected for EIGRP` does not work. To solve this problem, we chose to set EIGRP metrics manually, with the command: `redistribute bgp 65001 include-connected 10000 100 255 1 1500`, making the AS that uses EIGRP able to learn BGP routes.

Another issue that occurred in the lab was setting up IPv6 for BGP, on router 4 we issued the command: `neighbor 2001:DB8:ACAD:4::2 remote-as 65003`, while on router 5, we issued the command: `neighbor 2001:DB8:ACAD:4::1 remote-as 65001`. This incorrect configuration of the ASU resulted in the PuTTY terminal being spammed with debug messages and a redistribution between router 4 and 5, this was solved after we shut down the interfaces and reconfigured the ASNs to the correct ones.

Conclusion:

In conclusion, this lab demonstrates how to set up a network with 3 different ASes using 3 different IGPs: OSPF IS-IS and EIGRP, with their routes shared across the network with the use of BGP. BGP is essential to master since BGP is arguably one of, if not the most important exterior gateway protocols, since it is the only protocol used to exchange information between different autonomous systems, the key to enabling global internet traffic.